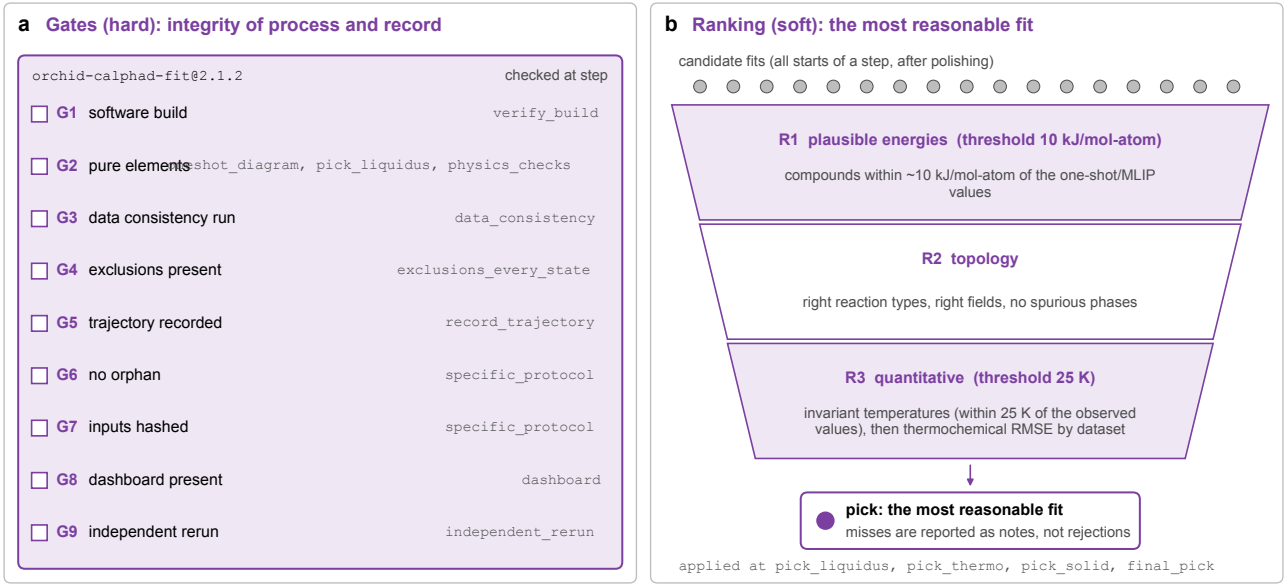

a Gates (hard): integrity of process and record
orchid-calphad-fit@2.1.2
checked at step
G1 software build
verify_build
G2 pure elements
one_shot_diagram, pick_liquidus, physics_checks
G3 data consistency run
data_consistency
G4 exclusions present
exclusions_every_state
G5 trajectory recorded
record_trajectory
G6 no orphan
specific_protocol
G7 inputs hashed
specific_protocol
G8 dashboard present
dashboard
G9 independent rerun
independent_rerun
b Ranking (soft): the most reasonable fit
candidate fits (all starts of a step, after polishing)
R1 plausible energies (threshold 10 kJ/mol-atom)
compounds within ~10 kJ/mol-atom of the one-shot/MLIP values
R2 topology
right reaction types, right fields, no spurious phases
R3 quantitative (threshold 25 K)
invariant temperatures (within 25 K of the observed values), then thermochemical RMSE by dataset
pick: the most reasonable fit
misses are reported as notes, not rejections
applied at pick_liquidus, pick_thermo, pick_solid, final_pick